

SFWE 403 - Clinical Laboratory Management System, University of Arizona

System ConOps and Requirements

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1 Project Overview

The Clinical Laboratory Management System (CLMS) is an integrated software platform designed to manage patient test requests, sample collection, diagnostic workflows, and result reporting. Unlike a general laboratory system, a clinical lab must capture the ordering physician's details and patient health insurance information for each test request, ensuring accurate medical accountability and billing.

The CLMS also supports the integration of multiple diagnostic machines and instruments (e.g., hematology analyzers, blood chemistry systems, microbiology platforms), enabling direct communication between the instruments and the system for seamless result entry, quality control, and equipment tracking.

2 Objectives

The objectives of a Laboratory Management System include:

1. **Streamlined Sample and Test Workflow:** To manage test orders, sample collection, processing, and reporting in a unified, traceable system.
2. **Accuracy in Diagnostics:** To minimize manual errors by automating test entry, data validation, and result generation.
3. **Quality and Patient Safety:** To enforce quality control checks, flag critical results, and ensure patient safety through standardized diagnostic protocols.
4. **Regulatory and Accreditation Compliance:** To comply with healthcare regulations and accreditation bodies (e.g., CLIA, CAP, ISO 15189), with proper audit trails and documentation.
5. **Efficient Billing and Insurance Handling:** To process laboratory billing, manage insurance claims, and track payments with precision.
6. **Improved Patient and Provider Communication:** To ensure timely delivery of test results to patients and referring physicians via secure channels.
7. **Comprehensive Analytics and Reporting:** To provide insights on test volumes, turnaround times, resource consumption, and staff performance.
8. **Data Security and Confidentiality:** To safeguard patient and laboratory data in compliance with healthcare privacy regulations such as HIPAA and GDPR.
9. **Physician & Insurance Capture:** Ensure that each test request is linked to the prescribing physician and the patient's insurance details for compliance, billing, and traceability.
10. **Instrument Integration:** Provide secure, automated connections to analyzers and diagnostic equipment for accurate result transmission.

3 Stakeholders

- **Healthcare Providers (Doctors, Nurses):** Referring physicians and health providers who submit test requests and use results for patient diagnosis and treatment planning. Interact with the system indirectly by submitting test orders and medical information that the laboratory processes.
- **Patients:** End-users who initiate the request for a clinical test, benefiting from improved accuracy and efficiency. They provide biological samples and receive test results for diagnostic purposes.

- **Phlebotomists and other Specimen collectors:** Work directly with the patient, to collect a sample for laboratory tests. They rely on the CLMS to print the barcode that identify every sample and tracks them.
- **Laboratory Technicians:** Operate machines such as blood analyzers, chemistry platforms, and microbiology systems, to test the samples collected by the phlebotomist. The test results should be integrated into the CLMS. They also rely on the CLMS to manage the inventory of reagents.
- **Laboratory Owners/Managers:** Oversee the overall operations, financial management, and regulatory compliance, using the system for reporting and decision-making.
- **Regulatory and Accreditation Authorities:** Require standardized reporting and quality assurance records and ensure that the laboratory adheres to laws and regulations, often requiring access to reports and data generated by the system.
- **Insurance Companies and Billing Partners:** Interact with the system for processing claims and verifying patient coverage, ensuring accurate billing and reimbursement.

4 Capabilities

- **Test and Sample Management**
 - **Order Entry:** Physicians or front-desk staff can easily enter and verify test requests, ensuring accurate collection of the sample.
 - **Sample Tracking:** Unique identifiers for every sample, ensuring complete traceability from collection to reporting.
 - **Critical Result Alerts:** Automatic flagging and notifications for life-threatening findings.
 - **Result Validation:** Results should be approved first by a technician and then by a pathologist before sending them to the patient and referring physician.
 - **Physician Information Requirement:** The system will require the name, license/ID, and contact details of the ordering physician for every test.
 - **Insurance Information Requirement:** Each test request must include patient insurance provider, policy number, and coverage details.
 - **Machine Integration:** Connects to laboratory instruments (e.g., hematology, blood chemistry, immunology analyzers) to automatically capture results.
- **Inventory and Equipment Management**
 - **Reagent and Consumable Tracking:** Real-time monitoring of quantities, expiration dates, and stock levels of reagents, test kits, and consumables to prevent shortages and overstocking.
 - **Automatic Reordering:** Generates purchase orders when stock falls below predefined levels, ensuring timely replenishment.
 - **Batch and Lot Management:** Tracks reagents, test kits, and consumables by batch and lot numbers, facilitating recalls and expiration tracking.
 - **Equipment Maintenance Management:** Tracks calibration, preventive maintenance, and equipment downtime logs.
- **Billing**
 - **Point of Sale (POS):** Integrated POS system for quick and accurate billing, including support for multiple payment methods.

- **Insurance Claim Processing:** Streamlines insurance billing, with automated claim generation and submission, and real-time status updates.
- **Patient and Provider Management**
 - **Patient Profiles:** Maintains comprehensive patient records, including personal details, medical history, and medication history.
 - **Provider Integration:** Links test requests and reports directly to referring physicians or hospitals.
 - **Medication History:** Provides a detailed history of all medications dispensed to each patient, helping in monitoring and follow-up.
 - **Tests Tracking:** Updates the patient records with each sample and test results, ensuring complete traceability from collection to reporting.
 - **Test History:** Provides a detailed history of all test results to each patient, helping in monitoring and follow-up.
 - **Report Delivery:** Results can be securely shared via patient portals, email, or physician dashboards.
- **Reporting and Analytics**
 - **Sales Reports:** Generates detailed sales reports, including daily, weekly, and monthly summaries.
 - **Inventory Reports:** Provides insights into stock levels, turnover rates, and drug usage patterns.
 - **Compliance Reports:** Generates reports required for regulatory compliance, such as a disease spreading report for the CDC and inventory audits.
- **Regulatory Compliance**
 - **Controlled Substances Management:** Tracks and manages controlled substances in compliance with legal requirements.
 - **Audit Trails:** Maintains detailed logs of all transactions and changes within the system, supporting audits and investigations.
 - **Data Security and Privacy:** Ensures compliance with data protection regulations (e.g., HIPAA) by securing patient data and limiting access.
- **User Management**
 - **Role-Based Access Control (RBAC):** Assigns roles and permissions to different users, ensuring that access to sensitive functions and data is restricted to authorized personnel.
 - **User Activity Monitoring:** Tracks and logs user activities within the system, helping in identifying and addressing any unauthorized actions.
- **Integration Capabilities**
 - **Third-Party Integration:** Integrates with other healthcare systems, such as Electronic Health Records (EHR), Laboratory Management Systems (PMS), and accounting software.
 - **Laboratory Network Integration:** Supports integration across multiple branches, enabling centralized management and reporting.

5 System Architecture

5.1 Overview of the System Architecture

The system should be built on client-server architecture, with a robust backend database supporting the application logic and a user-friendly graphical user interface (web-based or

desktop application) for ease of use. To support future expansion and scalability to multi-laboratory locations, the system *may* also include cloud-based options for scalability and remote access, ensuring continuous availability and easy maintenance.

The user interface is designed to be intuitive and accessible, with clear navigation, customizable dashboards, and user-friendly forms. It currently supports only the English language functionality and is optimized for desktop devices, although it should also be scalable to a multi-platform compatibility in the future allowing specimen collectorss and technicians to access the system on the go.

6 System Requirements

6.1 General System Requirements

6.1.1 Info Storage

The system shall store basic laboratory information, including name, website, address, owner, phone number and working hours.

6.1.2 GUI

The system shall provide an easy-to-use graphical user interface (GUI) for all interactions between the laboratory system and the laboratory staff.

6.1.3 User Types

The system shall support the following user types: laboratory manager, specimen collector, laboratory technician, cashier, patients, and healthcare provider.

6.1.4 Patient Accounts

The system shall allow all laboratory personnel to create and/or update patient accounts and store the following patient information: name, date of birth, address, phone number, email address and insurance information.

6.1.5 Patient Deletion

The system shall allow only a specimen collectors or laboratory manager to remove a patient from the system.

6.2 User Log-In Requirements

6.2.1 Account creation

The system shall allow only the laboratory manager to create new laboratory staff user accounts.

6.2.2 User Log In

The system shall allow the user to log in using the laboratory manager-created user account.

6.2.3 Create Password

The system shall require that the user create a unique password the first time they are logging into the system.

6.2.4 Change Password

The system shall allow an authorized user to change the password on their user account.

6.2.5 User Authentication

The system shall provide user authentication upon sign on for all external users (patients and health care providers) and laboratory staff including managers, specimen collectors, laboratory technicians and cashiers.

6.2.6 User Lock Out

The system shall lock-out any user that attempts to log in without proper authentication after 5 sequential attempts and send an email to the laboratory manager when this occurs.

6.2.7 Lock Out Recovery

The system shall only allow the laboratory manager to recover/unlock a locked-out account for a given user.

6.3 Test System Requirements

6.3.1 Accept Tests

The system shall allow all laboratory personnel to accept and enter Tests into the system that are brought directly to the laboratory by patients.

6.3.2 Manual Test Entry

The system shall allow laboratory staff to manually enter Tests received from a doctor's office via telephone.

Note added on Oct 1st:

Every test must have the following fields:

Report Information (i.e., Report ID and Report Date & Time),
Patient Information (i.e., Patient ID, Patient Name, Date of Birth / Age, and Gender),
Ordering Physician Information (i.e. Physician ID and Physician Name),
Test Execution Details (i.e. Sample Type (e.g., blood, urine or tissue), Sample ID,
Collection Date & Time, Collected By (technician ID/name), Test Performed By
(technician/scientist ID), and Machine/Instrument Used),
Test Results (i.e. Test Name, Result Values (quantitative/qualitative), Units, Normal
Reference Range, and Result Status (pending, completed, validated), Results
Received Date & Time), and
Health Insurance / Billing (i.e. Insurance Provider and Insurance Policy Number)

6.3.3 Reagent Availability

The system shall allow only specimen collectors to check the availability of reagents, test kits, or consumables in inventory prior to collecting a sample.

6.3.4 Collecting Samples

The system shall allow only specimen collectors to print the barcode for sample collection one time.

6.3.5 Test Tracking

The system shall keep track of all samples collected/tested for a given patient and make it available to all laboratory personnel.

Note added on Oct 1st:

To keep track of tested samples, the system shall read the files produced by the laboratory machines.

Every machine output the test result as a plain-text file (.txt) using the following simplified version of the ORU - HL7 message:

BARCODE|Message type (Test)^Test ID^Test description|Value|Date and time

Examples:

1234|Test^B123^T3 Uptake|28|202208221340

5678|Test^U123^Nitrite|Absent|202208221340

6.3.6 Activity Log

The system will keep an activity log (i.e., specimen collectors name, sample number, patient name, date and time of collection, type of test, machine ID, date and time of testing) for all tests filled.

6.3.7 Expired Chemicals

The system shall display a warning and prevent testing samples with expired reagents, test kits, or consumables.

6.4 Inventory Control System Requirements

6.4.1 Inventory Tracking

The system shall update the inventory when a specimen collector fills a test.

6.4.2 Low Stock

The system shall notify the laboratory manager about machines with low stock (less than 20% of reagents, test kits, or consumables).

6.4.3 New Chemical Orders

The system shall allow only the laboratory manager to order new reagents, test kits, or consumables when quantities run low.

6.4.4 New Supply Received

The system shall update the inventory whenever new reagents, test kits, or consumables are received from a distributor.

6.4.5 Update Inventory

The system shall allow the laboratory manager to update the inventory when new reagents, test kits, or consumables are used.

6.4.6 Expired Chemical Notification

The system shall notify the laboratory manager about expired reagents, test kits, or consumables that will expire within 30 days.

6.4.7 Remove Inventory

The system shall allow the laboratory manager to remove inventory from the system when chemicals have expired or other reasons and record the removal in an activity log.

6.5 Reporting System Requirements

6.5.1 Transaction logging

The system shall log all transactions, such as logins, logouts, purchases, and inventory updates.

6.5.2 Financial Reports

The system shall allow laboratory managers to generate financial reports and statistics for a specified period.

6.5.3 Inventory Reports

The system shall allow laboratory managers to generate inventory reports for a specified period of time.

6.5.4 Operational Reports

The system shall allow laboratory technicians measure turnaround times, machine performance, and machine maintenance.

6.6 Purchasing System Requirements

6.6.1 Process Test Purchases

The system shall allow the laboratory staff to process Test items for purchase via manual input from the GUI.

6.6.2 POS Signature

The system shall require that customers sign (does not have to be electronic signature) for all taken samples at the point of sale.

6.6.3 Purchase Receipts

The system shall generate and print receipts for the entire customer purchase.

6.6.4 Payment Types

The system shall support multiple payment methods, including cash, debit, and credit cards.

7 Change Requests

7.1 Change Request Template

Date:

Team Number:

Project Name:

Title of Change: Enter a descriptive phrase for reference

Documents Affected (check all that apply):

	Doc Type	Revision	Document / File Name
	SRS		
	<i>Source Code</i>		
	<i>Test Procedures</i>		

Description of Change: What is it that needs to be changed?

Need for Change: Why must this change be made?

Impact: How will this change impact your project if it is, or is not implemented?

Substantiation: What have you done to verify that this change will work?

Approvals:

Team Leader Signature

Instructor Signature